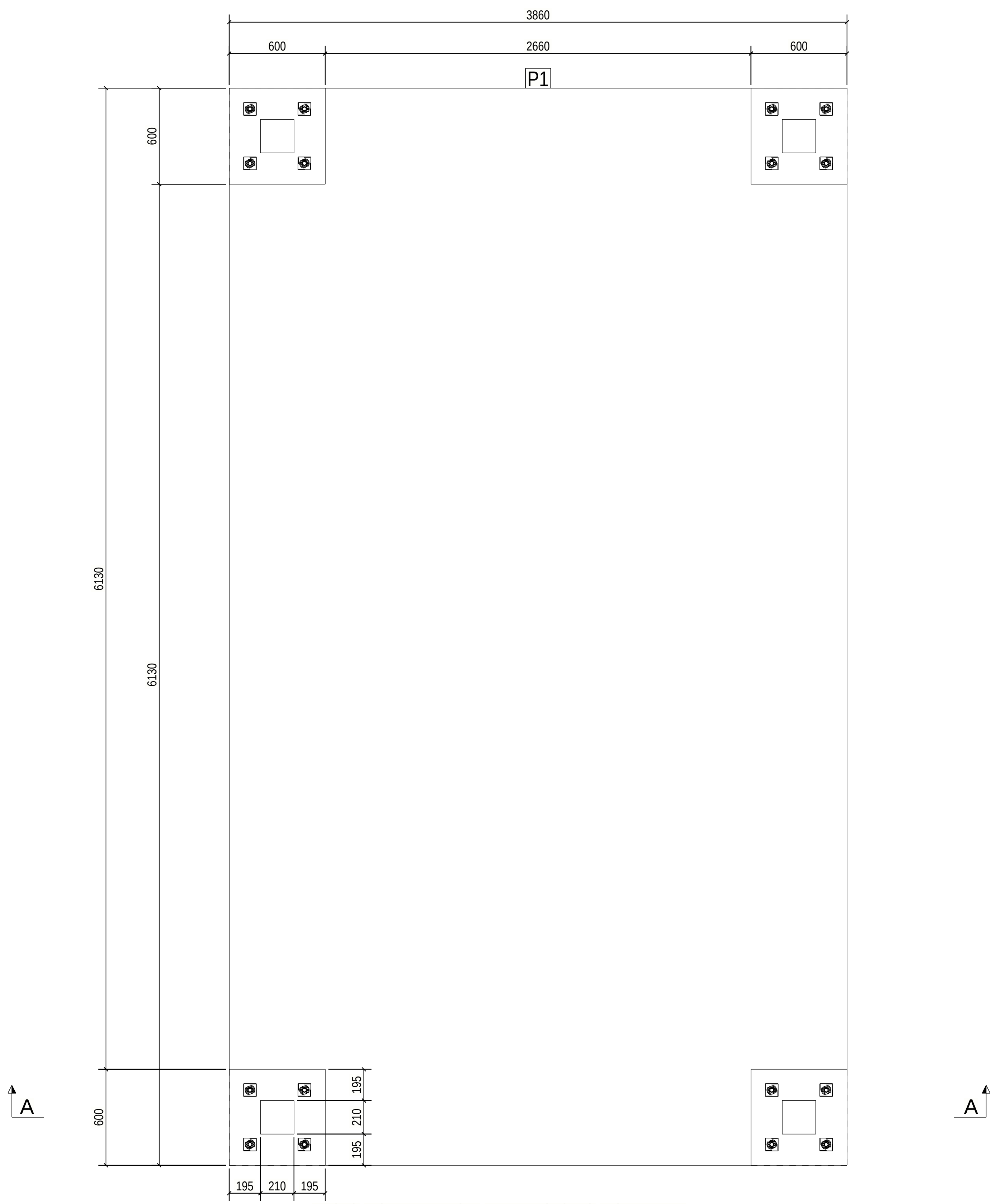
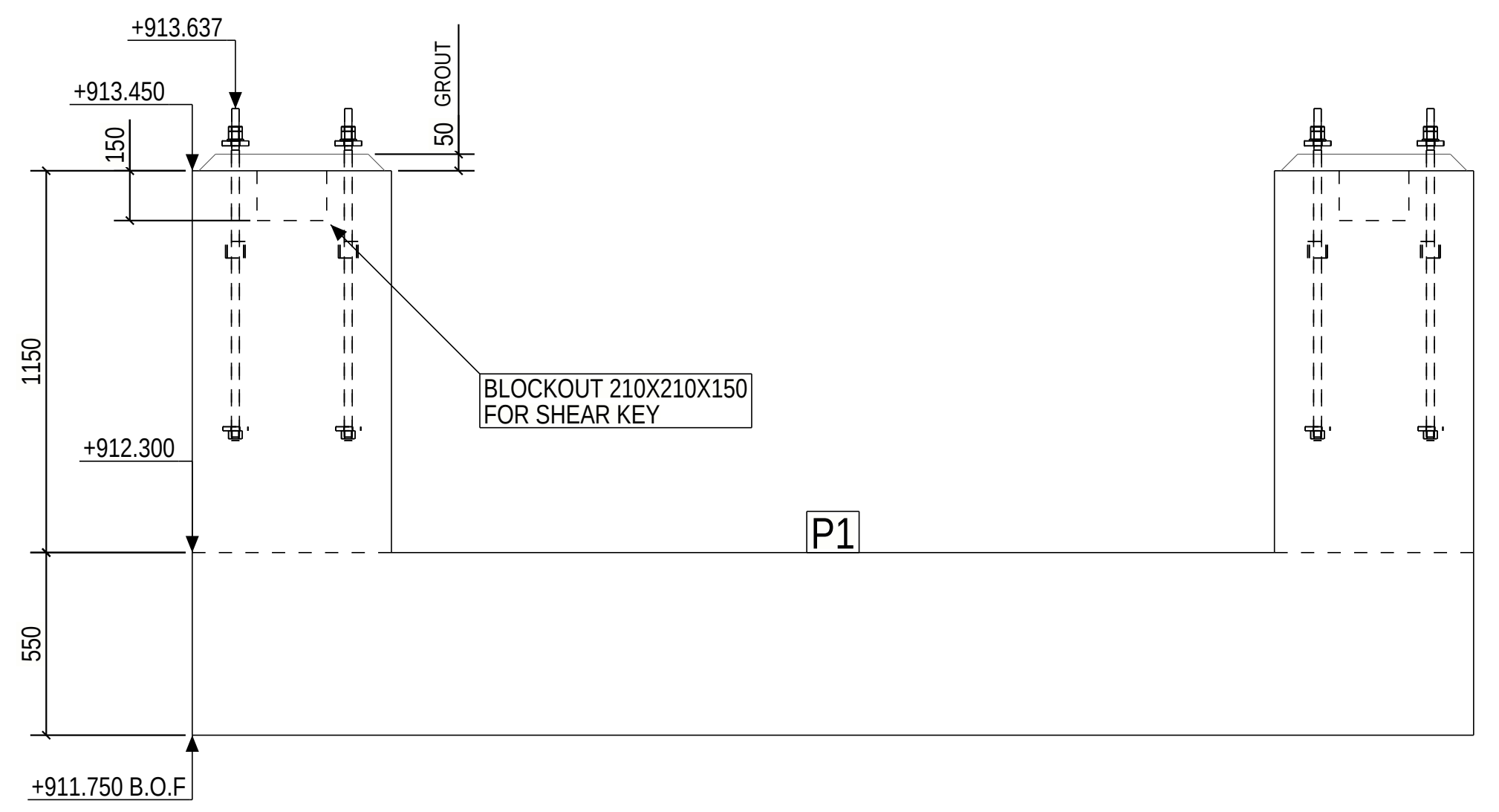
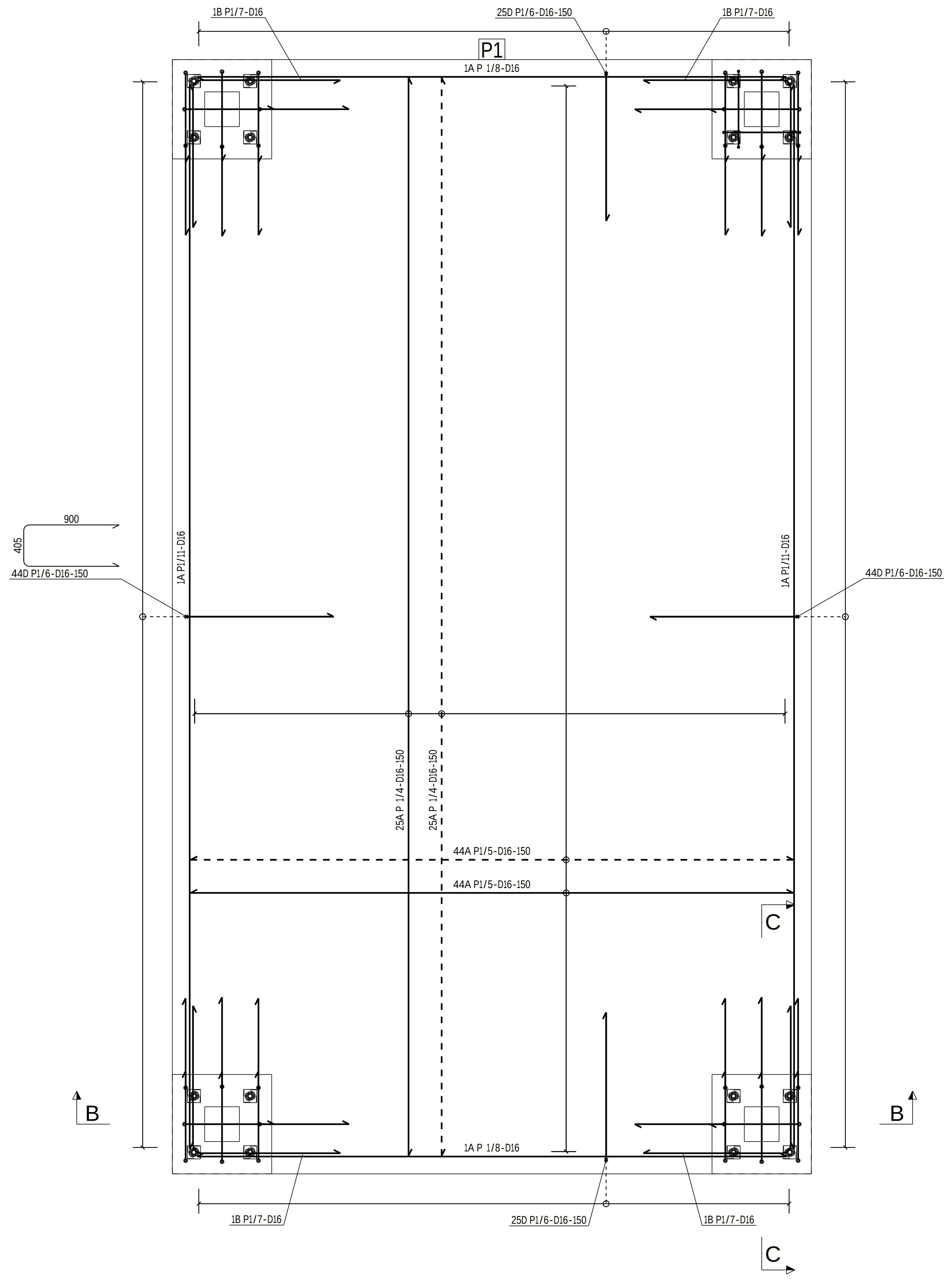
MD-216-49W1-EG-CV-DED-1006
DRAWING NUMBER
D01
REV


3860
600
2660
600
P1
600
6130
6130
600
195
210
195
A
A

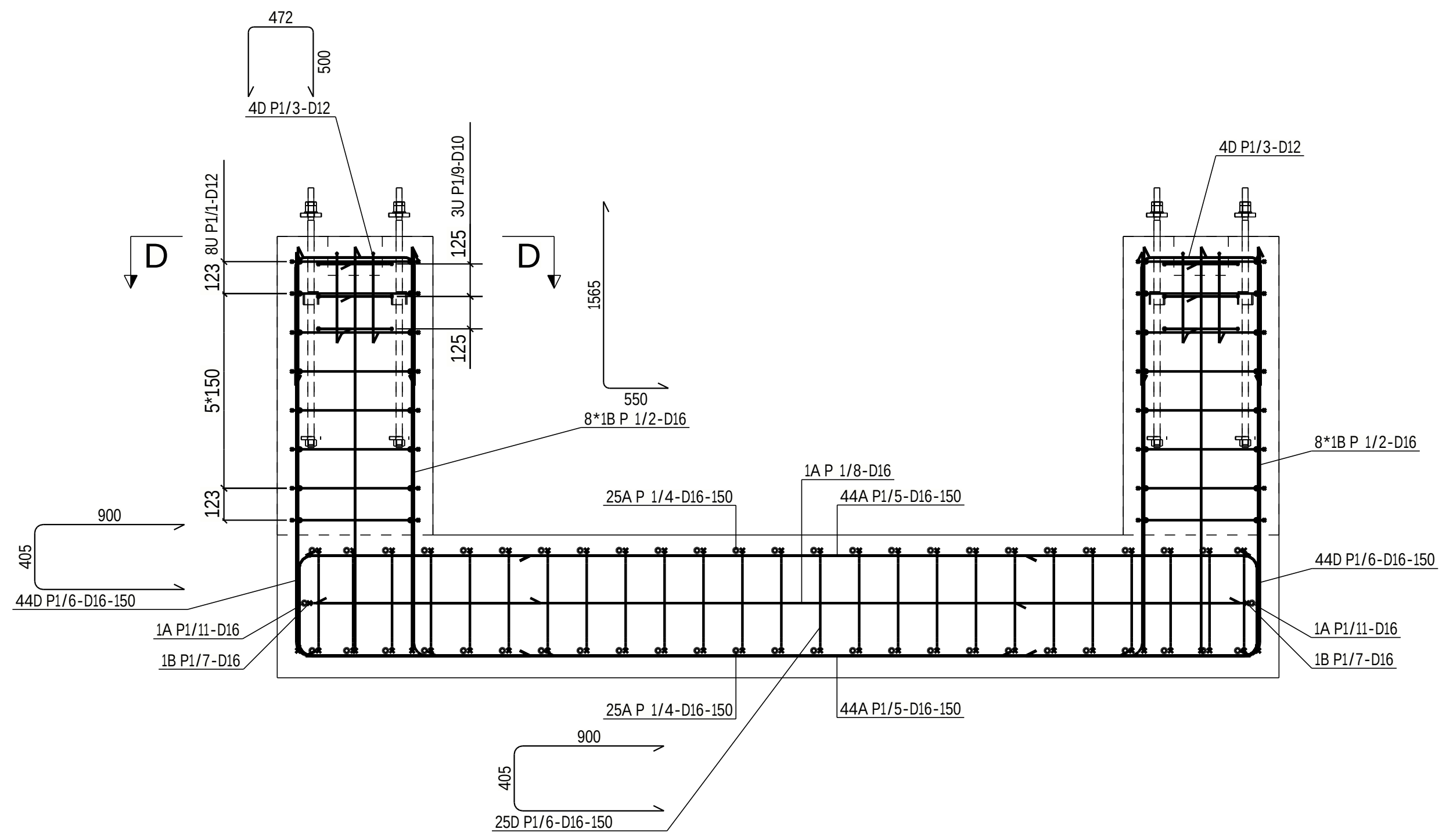
R.C.C FOUNDATION DIMENSIONS TOP VIEW
(SCALE 1:20)


+913.637
+913.450
150
50
GROUT
1150
+912.300
BLOCKOUT 210X210X150
FOR SHEAR KEY
P1
550
+911.750 B.O.F

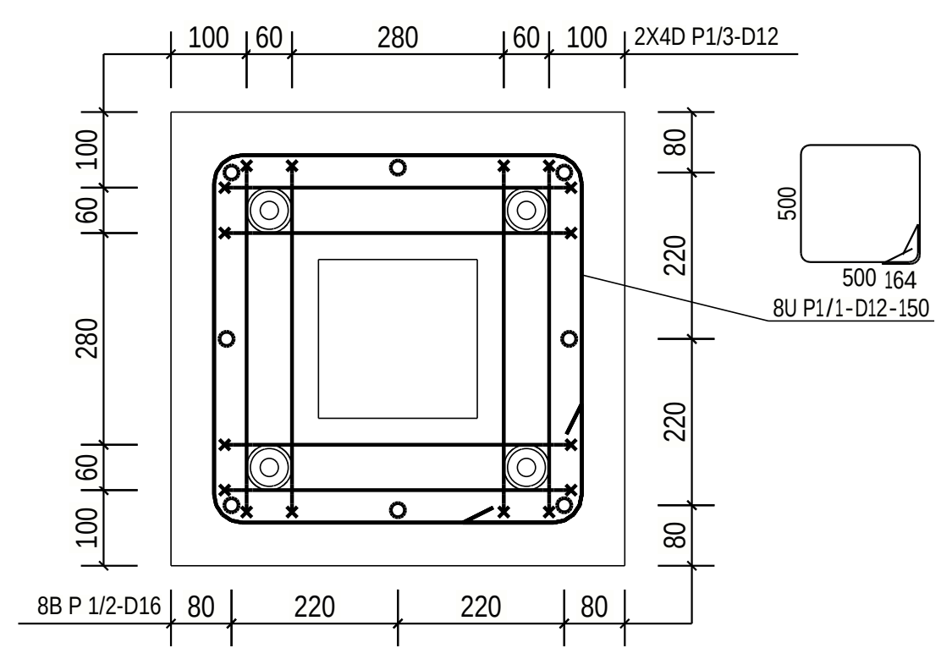
SECTION : A - A
(SCALE 1:20)


1B P1/7-D16
25D P1/6-D16-150
1B P1/7-D16
P1
1A P 1/8-D16
900
405
44D P1/6-D16-150
1A P1/11-D16
1A P1/11-D16
44D P1/6-D16-150
25A P 1/4-D16-150
25A P 1/4-D16-150
44A P1/5-D16-150
44A P1/5-D16-150
C
B
B
1A P 1/8-D16
1B P1/7-D16
25D P1/6-D16-150
1B P1/7-D16
C

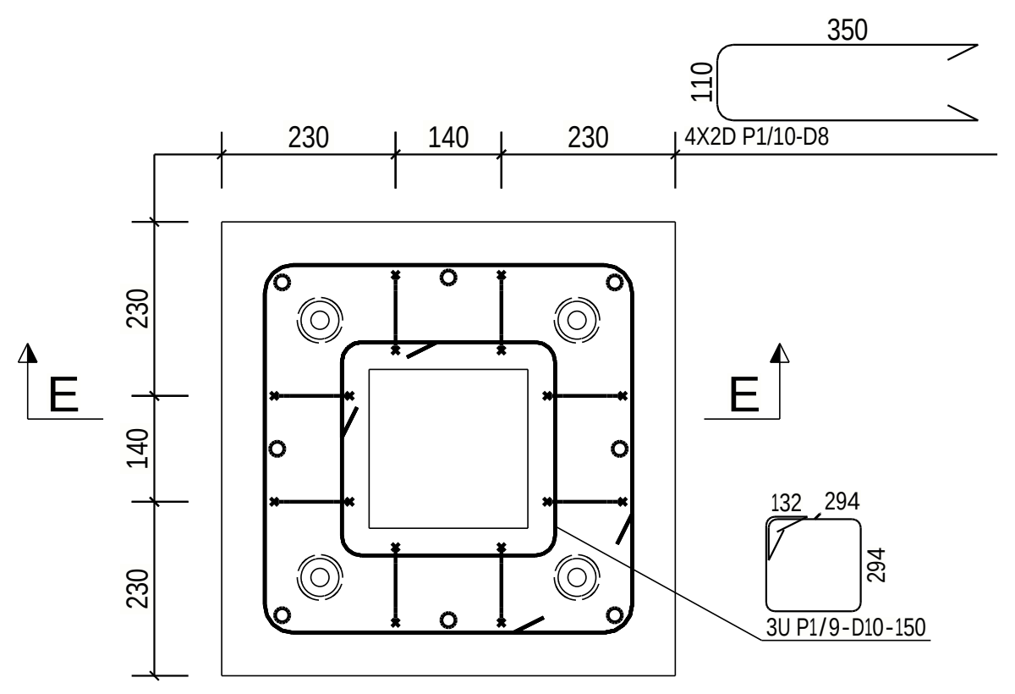
R.C.C FOUNDATION REINFORCEMENT TOP VIEW
(SCALE 1:20)


472
500
4D P1/3-D12
8U P1/1-D12
3U P1/9-D10
D
D
123
125
125
5*150
123
1565
550
8*1B P 1/2-D16
4D P1/3-D12
8*1B P 1/2-D16
1A P 1/8-D16
25A P 1/4-D16-150
44A P1/5-D16-150
900
405
44D P1/6-D16-150
44D P1/6-D16-150
1A P1/11-D16
1B P1/7-D16
1A P1/11-D16
1B P1/7-D16
25A P 1/4-D16-150
44A P1/5-D16-150
900
405
25D P1/6-D16-150

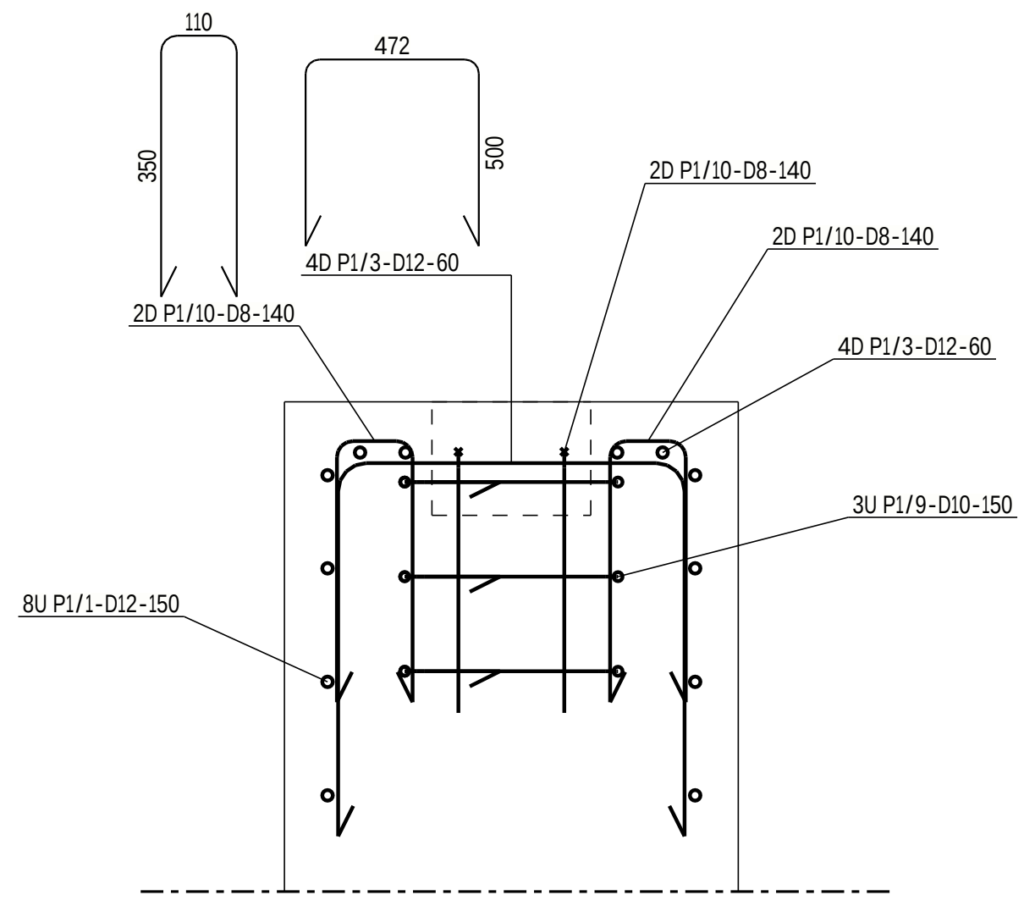
SECTION : B - B
(SCALE 1:20)


100
60
280
60
100
2X4D P1/3-D12
100
60
280
60
100
80
220
220
80
500
500 164
8U P1/1-D12-150
8B P 1/2-D16
80
220
220
80

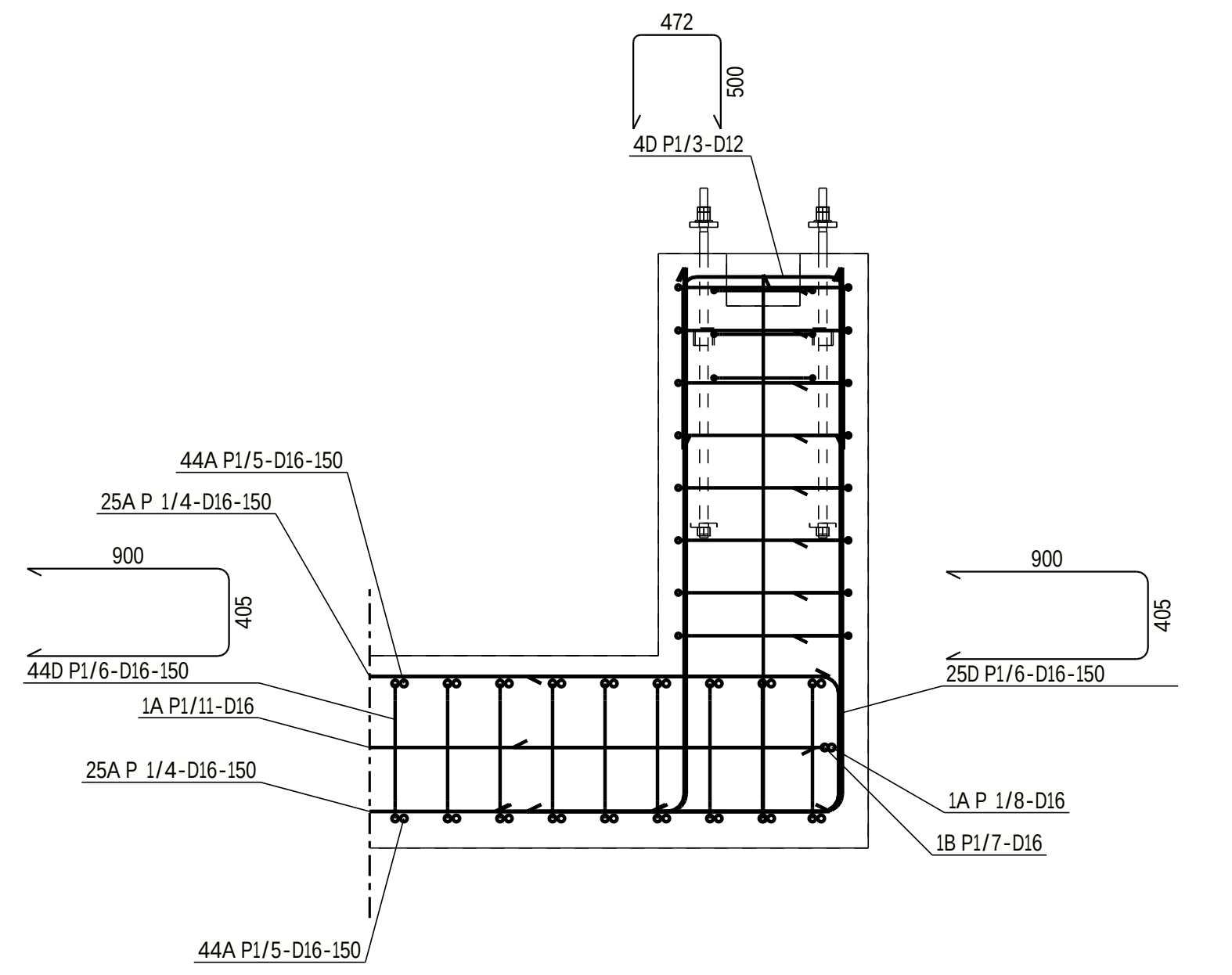
SECTION : D - D MAIN TOP REINFORCEMENT
(SCALE 1:10)


350
110
230
140
230
4X2D P1/10-D8
230
140
230
E
E
132
294
294
3U P1/9-D10-150

SECTION : D - D EXTRA TOP REINFORCEMENT
(SCALE 1:10)


110
350
472
500
4D P1/3-D12-60
2D P1/10-D8-140
2D P1/10-D8-140
2D P1/10-D8-140
4D P1/3-D12-60
3U P1/9-D10-150
8U P1/1-D12-150

SECTION : E - E
(SCALE 1:10)


472
500
4D P1/3-D12
44A P1/5-D16-150
25A P 1/4-D16-150
900
405
44D P1/6-D16-150
1A P1/11-D16
25A P 1/4-D16-150
900
405
25D P1/6-D16-150
1A P 1/8-D16
1B P1/7-D16
44A P1/5-D16-150

SECTION : C - C
(SCALE 1:20)

CONCRETE INFORMATION

Concretes used	C35	
Concrete cover 1	75±10mm	BOTTOM AND SIDE OF FOUNDATION
Concrete cover 2	50±10mm	TOP OF FOUNDATION, COLUMN

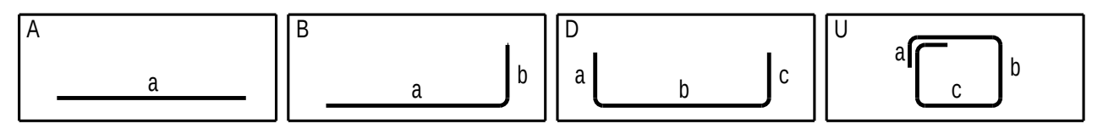
REINFORCING BAR LIST

CAST UNIT CODE: P1 PCS: 1

REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
U	P1/1	32	B500B	12	2170		61.9	164	500	500	500	500				60	
B	P1/2	32	B500B	16	2070		104.9	550	1565				90			80	
D	P1/3	32	B500B	12	1410		40.1	500	472	500						60	
A	P1/4	50	B500B	16	6510		514.3	6510									
A	P1/5	88	B500B	16	3640		506.1	3640									
D	P1/6	138	B500B	16	2120		463.2	900	405	900						80	
B	P1/7	4	B500B	16	1750		11.1	900	900				90			80	
A	P1/8	2	B500B	16	3560		11.2	3560									
U	P1/9	12	B500B	10	1320		9.8	132	294	294	294	294				40	
D	P1/10	32	B500B	8	770		9.8	350	110	350						32	
A	P1/11	2	B500B	16	6430		20.3	6430									

REINFORCING BAR TOTAL WEIGHT [kg] 1752.7

REINFORCEMENT BENDING SHAPES:


A
a
B
a
b
D
a
b
c
U
a
b
c

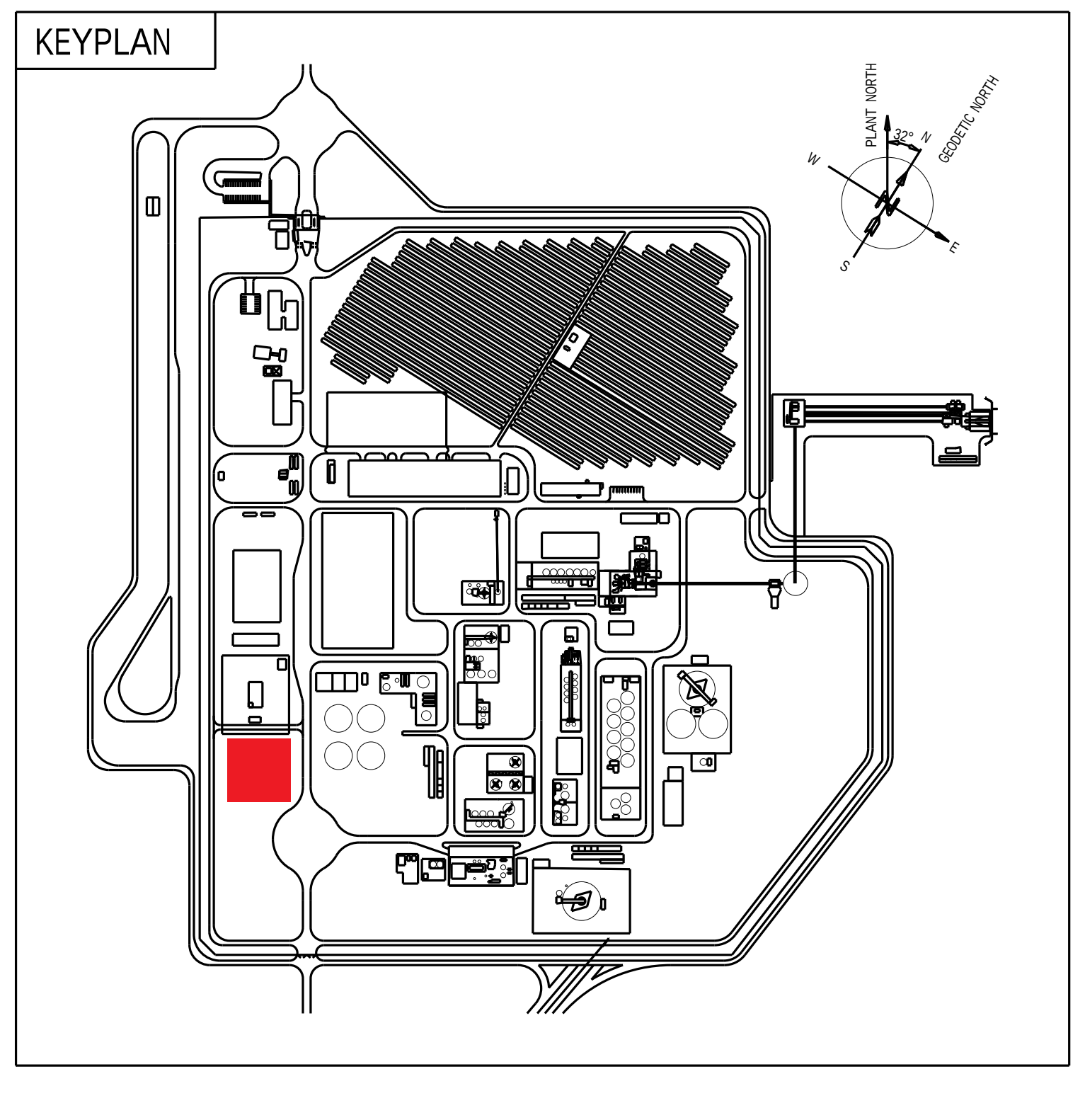
LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 UNLESS NOTED OTHERWISE
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DEEP GALVANIZED
8. NET BEARING CAPACITY UNDER FOUNDATION 375 kPa

KEYPLAN


PLANT NORTH
GEODETIC NORTH
32°
N
W

DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-GAD-1003	1-10/A-E GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-CAL-1005	STAIRCASE FOUNDATIONS CALCULATION REPORT
MD-216-49W1-EG-CV-GAD-1001	1-8/E-J GENERAL ARRANGEMENT DRAWING	MD-216-49W1-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NanWu	LuiPiz	
C01	23.11.2021	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NanWu	LuiPiz	
A01	05.11.2021	ISSUED FOR APPROVAL	JonRen	MauNot	NanWu	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_D_24892_006 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_CTB01_49006 — DOCUMENT ID: OU100732638 — REVISION: 2

DRAWN	JonRen	29.10.2021
CHECKED	MauNot	29.10.2021
ENGINEER	NanWu	29.10.2021
APPROVED	LuiPiz	29.10.2021
EMPLOYER		
SCALE	1:20	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
GL 4-6/H-J STAIRCASE FOUNDATIONS
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER: MD-216-49W1-EG-CV-DED-1006
REV: D01